

# FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 — SUB-STRAND 2.5 RATES OF REACTIONS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Welcome to your Final Explanation! On Day 1 you watched the Disappearing Thiosulfate Cross: two beakers with the same sodium thiosulfate and dilute hydrochloric acid, one in cold water (about 10 °C, like a cool Nairobi highland morning) and one in warm water (about 50 °C, like a hot Mombasa or Kisumu afternoon). The cross under the warm beaker disappeared in about 30-40 seconds, while the cross under the cold beaker stayed visible for several minutes. Now you will use everything you learned in Lessons 1-8 to explain what happened. Write your answers in the space under each prompt. Use particle ideas (collisions, energy, activation energy) and evidence from our experiments. Use full sentences, and label any diagrams you draw. Think about our driving question: Why do some reactions happen in a flash while others take days, and how can we control their speed in Kenya?

### Part 1: Explaining the Disappearing Cross

**Look back at the two beakers from Day 1. Both held the same chemicals in the same amounts. (a) How did you measure the speed of the reaction, and what is meant by 'rate of reaction'? (b) Use ideas about particles, collisions and energy to explain why the cross under the warm beaker (about 50 °C) disappeared in about 30-40 seconds while the cross under the cold beaker (about 10 °C) stayed visible for several minutes. Include a labelled particle sketch of both beakers.**

(a) We timed how long it took for the black cross to disappear as sulfur precipitate formed. Rate of reaction is how quickly reactants are used up or products are formed over time. For this method the rate can be written as  $1/t$  ( $s^{-1}$ ), so a shorter time means a higher rate. (b) The chemicals and amounts were identical, so the only difference was temperature. In the warm beaker (about 50 °C) the sodium thiosulfate and HCl particles have more kinetic energy. They move faster, so they collide more frequently, and each collision carries more energy. This means a greater proportion of collisions have energy equal to or greater than the activation energy, so more collisions are successful and sulfur forms faster. The cross disappeared in about 30-40 seconds. In the cold beaker (about 10 °C) the particles move more slowly, collide less often and with less energy, so fewer collisions are successful and the cross stayed visible for several minutes. Sketch: two beakers, warm with particles drawn with long motion arrows and many collisions, cold with short arrows and few collisions, each labelled with its temperature and rate (fast or slow).

### Part 2: Concentration and Surface Area

**In Lessons 3-5 you investigated concentration (diluting sodium thiosulfate to 100%, 75%, 50% and 25%) and surface area**

(a) Concentration: as the sodium thiosulfate became more dilute, the time for the cross to disappear increased. The full-strength solution was fastest and the 25% solution was slowest. Plotting rate ( $1/\text{time}$ ) against concentration showed rate increasing as concentration increased. Surface area: marble powder reacted fastest, small chips were slower, and large chips were slowest, shown by rapid fizzing and faster mass loss for the powder, with slow steady bubbling for large chips. (b) Higher concentration

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| <p>(large marble chips, small chips and powder with dilute HCl). (a) Describe the pattern you found for each factor, with the evidence that supports it. (b) Explain each pattern using collision ideas. (c) In the marble experiment all three flasks finally lost the same total mass. What does this tell you about what surface area changes and what it does not change?</p> | <p>means more reactant particles in the same volume, so particles collide more frequently and more successful collisions happen each second. Particle speed stayed the same; only the number of particles per unit volume changed. For a solid, crushing exposes more particles of marble (<math>\text{CaCO}_3</math>) to the HCl particles at any moment, so there are more collisions per second and <math>\text{CO}_2</math> forms faster. (c) The same total mass loss shows that the same amount of marble reacted in the end. Surface area changes only how fast the reaction happens (the rate), not how much product forms in total. Surface area matters only for solid reactants.</p> |
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### Part 3: Catalysts and Light

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| <p>In Lessons 6 and 7 you tested fresh liver, boiled liver and sand with hydrogen peroxide, and you tested manganese dioxide (<math>\text{MnO}_2</math>) and light on reactions. (a) What did you observe in these tests, and what evidence showed that <math>\text{MnO}_2</math> is a catalyst? (b) Explain how a catalyst speeds up a reaction, and say how this is different from the way concentration and surface area work. (c) Why did boiled liver not work? (d) How does light speed up some reactions?</p> | <p>(a) Fresh liver and <math>\text{MnO}_2</math> both caused vigorous bubbling of oxygen with hydrogen peroxide (<math>\text{MnO}_2</math> relit a glowing splint). Boiled liver and sand caused no reaction. The <math>\text{MnO}_2</math> powder was still visible on the filter paper after the reaction, so it was not used up, which is the evidence that it is a catalyst. Photosensitive paper in bright light or sunlight changed colour faster than in dim light, and paper in the dark did not change. (b) A catalyst provides an alternative reaction pathway with a lower activation energy, so at the same temperature a greater fraction of collisions has enough energy to be successful. It is not consumed. Concentration and surface area instead increase how often particles collide. (c) Liver contains the enzyme catalase, a biological catalyst. Boiling destroys its three-dimensional shape, so it can no longer speed up the reaction. (d) Light supplies energy directly to reactant particles, increasing the proportion of collisions or particles with enough energy to react, so photochemical reactions happen faster in brighter light.</p> |
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### Part 4: Dagaa Drying at Lake Victoria

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| <p>A fisher at Lake Victoria wants to dry dagaa and stop it from spoiling quickly. Spoilage is caused by chemical and biological reactions. (a) Choose two rate factors you studied and say how you would change each one to slow down spoilage reactions in the dagaa before it is dried, or to control the</p> | <p>(a) and (b) Temperature: keep fresh dagaa cool, for example in shade or on ice. Lower temperature means particles and enzymes have less kinetic energy, so there are fewer and less energetic collisions and fewer collisions exceed the activation energy, so spoilage reactions are slower. Surface area/exposure: spread dagaa in a thin layer so it dries quickly in sunshine and heat, because removing water lowers the concentration of reactants and the water needed for reactions, so collisions between reacting particles become less frequent. (Light and heat from sunshine also supply energy, but the main aim is drying.) Another valid choice is to lower concentration of reactants by removing water. (c) A factor that speeds up one reaction can be wanted or unwanted. Heating githeri raises temperature so the food cooks faster, which is useful, but the same higher temperature would speed up spoilage if the dagaa stayed warm and wet. In the gut, enzymes act as catalysts at body temperature to digest ugali efficiently, which is useful, but the same kind of catalyst activity in spoiling food is unwanted. So we choose to increase or decrease each factor depending on whether we want the reaction faster or slower.</p> |
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| <p>reactions. (b) Explain each choice using collision theory. (c) Give one reason why a rate factor that is helpful in one Kenyan situation may be unhelpful in another, using an example such as cooking githeri or the stomach digesting ugali.</p> |  |
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| Part 5: Closing the Case                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <p><b>Write a final answer to our driving question: Why do some chemical reactions happen in a flash while others take days, and how can we control the speed of reactions to make life in Kenya safer, healthier and more productive? In your answer, (a) name all the rate factors we studied and say what each changes at the particle level, and (b) give two examples from Kenyan life where controlling reaction speed is useful.</b></p> | <p>(a) Reaction speed depends on how many particle collisions are successful each second. Temperature increases both how often particles collide and how much energy they have, so more collisions exceed the activation energy. Concentration increases the number of particles per unit volume, so collisions are more frequent. Surface area (for solids) exposes more particles, so collisions are more frequent. Catalysts, including enzymes, lower the activation energy, so more collisions are successful without being used up. Light supplies energy to reactant particles in some reactions, so more collisions are successful. Reactions that happen in a flash have many successful collisions each second; slow ones, such as rusting of mabati roofs, have very few. (b) Examples: grinding maize at the posho mill or crushing charcoal for the jiko increases surface area and speeds up reactions; enzymes in the gut digest ugali and githeri at body temperature; keeping food cool, or drying dagaa, slows spoilage; and fertiliser made with an iron catalyst helps Kenyan farmers grow more food. By understanding and controlling temperature, concentration, surface area, catalysts and light, we can speed up helpful reactions and slow down harmful ones, so life is safer, healthier and more productive. The disappearing cross showed this: the same chemicals reacted at different speeds because warm particles collided more often and with more energy.</p> |

| RUBRIC                                     |                                                                                                                                                                 |                                                                                                                        |                                                                                                                              |
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| Criterion                                  | Excellent (4)                                                                                                                                                   | Proficient (3)                                                                                                         | Developing (1–2)                                                                                                             |
| Use of collision theory and particle ideas | Consistently explains rate differences using collision frequency, collision energy and activation energy, with clear labelled particle sketches.                | Uses collision ideas correctly in most answers but may omit energy or activation energy, or sketches lack some labels. | Gives descriptions of what happened with little or no particle-level explanation, or confuses speed and number of particles. |
| Use of evidence from the lessons           | Accurately uses observations from the thiosulfate cross, concentration, marble and catalyst experiments, including the 30-40 second and several-minute results. | Refers to most experiments with mostly accurate details.                                                               | Gives few or inaccurate references to the class experiments.                                                                 |
| Distinguishing between rate factors        | Clearly separates how temperature, concentration, surface area, catalysts and light each work, including that                                                   | Explains most factors correctly but with some overlap or missing detail.                                               | Treats the factors as the same or makes incorrect statements about them.                                                     |

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|                                | surface area changes rate but not total product and that catalysts are not consumed.                                                                                      |                                                               |                                                             |
| Application to Kenyan contexts | Applies the model accurately and creatively to dagaa drying, githeri, ugali digestion, posho milling or similar, and recognises that factors can be helpful or unhelpful. | Gives relevant Kenyan examples with a reasonable explanation. | Examples are missing, unclear or not linked to the science. |
| Scientific communication       | Writes clear, well-organised answers using correct terms such as rate, activation energy, catalyst and enzyme.                                                            | Writes understandable answers with mostly correct vocabulary. | Answers are unclear or use few scientific terms.            |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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